A First Course in Abstract Algebra

Section 28: Gröbner Bases for Ideals

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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28 Gröbner Bases for Ideals

Exercise 28.1: Decreasing lexicographic order, 1

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-3x^3 + 7x^2y^2z - 5x^2yz^3 + 2xy^3z^5.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.2: Decreasing lexicographic order, 2

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-4x + 5y^3z^3 + 3y^2z^5 - 8z^7.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.3: Decreasing lexicographic order, 3

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$2x^2yz^2 - 2xy^2z^2 - 7x + 3y + 10z^3.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.4: Decreasing lexicographic order, 4

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-8xy - 4xz + 3yz^3 + 2yz + 38.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.5: Reversed variable priority, 1

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 1 becomes

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

Exercise 28.6: Reversed variable priority, 2

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 2 becomes

$$-8z^7 + 3y^2z^5 + 5y^3z^3 - 4x.$$

Exercise 28.7: Reversed variable priority, 3

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 3 becomes

$$10z^3 - 2xy^2z^2 + 2x^2yz^2 + 3y - 7x.$$

Exercise 28.8: Reversed variable priority, 4

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 4 becomes

$$3yz^3 + 2yz - 4xz - 8xy + 38.$$

Exercise 28.9: The first twenty degree-lexicographic monomials

First sort by total degree, then by lexicographic order with $z < y < x$. There are $1, 3, 6, 10$ monomials of degrees $0, 1, 2, 3$, respectively, so these are exactly the first twenty:

$$1;$$
$$z,\ y,\ x;$$
$$z^2,\ yz,\ y^2,\ xz,\ xy,\ x^2;$$
$$z^3,\ yz^2,\ y^2z,\ y^3,\ xz^2,\ xyz,\ xy^2,\ x^2z,\ x^2y,\ x^3.$$

For instance $y^3 < xz^2$ despite the larger y exponent, because the latter has a positive x exponent.

Exercise 28.10: Decreasing degree-lexicographic order, 1

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.11: Decreasing degree-lexicographic order, 2

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3y^2z^5 - 8z^7 + 5y^3z^3 - 4x.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.12: Decreasing degree-lexicographic order, 3

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2x^2yz^2 - 2xy^2z^2 + 10z^3 - 7x + 3y.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.13: Decreasing degree-lexicographic order, 4

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3yz^3 - 8xy - 4xz + 2yz + 38.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.14: A one-step reduction of the largest leading monomial

The largest leading monomial is x^2y, in $f_2 = x^2y + 4xy$. It is divisible by the leading monomial xy of $f_3 = xy - y^2$. Replace

$$f_2 \longmapsto f_2 - xf_3 = xy^2 + 4xy.$$

The new basis is $\{xy^2 - 2x,\ xy^2 + 4xy,\ xy - y^2\}$. Its maximum monomial is xy^2, smaller than x^2y. The ideal is unchanged because the replacement can be reversed by adding xf_3.

Exercise 28.15: Reducing a mixed cubic basis

The first polynomial has leading monomial xy; the third, $x - y^4$, has leading monomial x. Replace the first by

$$(xy + y^3) - y(x - y^4) = y^5 + y^3.$$

The resulting basis $\{y^5 + y^3,\ y^3 + z,\ x - y^4\}$ generates the same ideal. Its largest monomial is now x, whereas the previous largest was xy. In pure lex order a monomial involving x exceeds every monomial involving only y, z, regardless of degree.

Exercise 28.16: When the maximum cannot be reduced in one step

The leading monomials are xyz, x^3, and x^2yz^3. The largest is x^3. Neither xyz nor x^2yz^3 divides x^3, because each contains positive y, z exponents. Thus no division step using another basis element can cancel that largest monomial. One can reduce the third polynomial using the first, but the unchanged x^3 still remains the maximum. Consequently a single-step reduction that strictly decreases the maximum monomial of the entire basis is impossible.

Exercise 28.17: A reduction involving only two variables

The largest leading monomial is y^3z^2, divisible by y^2z^2, the leading monomial of the third polynomial. Replace

$$(y^3z^2 - 2z) - y(y^2z^2 + 3) = -3y - 2z.$$

Thus $\{y^2z^3 + 3,\ -3y - 2z,\ y^2z^2 + 3\}$ generates the same ideal. The maximum has dropped from y^3z^2 to y^2z^3.

Exercise 28.18: A linear Groebner basis

Name the three displayed generators f_1, f_2, f_3 in their source order. Ordinary elimination produces

$$\begin{aligned} g_1 &= 3f_1 - f_3 = 2w + 11z - 4, \\ g_2 &= -7f_1 + 2f_2 + 3f_3 = 4x - 29z + 14, \\ g_3 &= -5f_1 + 2f_2 + f_3 = 4y - 23z + 18. \end{aligned}$$

The coefficient transformation has nonzero determinant -8, so it is invertible over $\mathbb{R}$: both triples generate the same ideal. The leading monomials are w, x, y, pairwise relatively prime. Their S-polynomials therefore reduce to zero, so $\{g_1, g_2, g_3\}$ is a Groebner basis. Equivalently this is row reduction of a linear system, with z free.

Exercise 28.19: An inconsistent linear ideal

Let the source generators be f_1, f_2, f_3. Subtract twice the first from the second to obtain $6x - 5y + 1$. Subtract the first from the third to obtain $-6x + 5y - 3$. Adding gives

$$f_2 + f_3 - 3f_1 = -2.$$

Thus the ideal contains the nonzero constant -2, hence also 1. It is the whole polynomial ring, and $\{1\}$ is a Groebner basis. Its algebraic variety is empty.

Exercise 28.20: A one-variable common divisor

Let $f = x^4 + x^3 - 3x^2 - 4x - 4$ and $g = x^3 + x^2 - 4x - 4$. The first division gives

$$f - xg = x^2 - 4, \qquad g = (x + 1)(x^2 - 4).$$

Therefore $(f, g) = (x^2 - 4)$. A single generator of a one-variable ideal is a Groebner basis: the leading term of every multiple is divisible by the generator's leading term. The answer is $\{x^2 - 4\}$.

Exercise 28.21: Three one-variable generators

For the first two generators f_1, f_2, division gives

$$f_1 - (x - 3)f_2 = 6(x^2 - 3x + 2), \qquad f_2 = (x + 2)(x^2 - 3x + 2).$$

Their gcd is $h = (x - 1)(x - 2)$. The third generator is $f_3 = (x - 1)^2(x + 2)$, and

$$f_3 - (x + 3)h = 4(x - 1).$$

All three original polynomials are divisible by $x - 1$, while the displayed steps put $x - 1$ in their ideal. Hence the ideal is $(x - 1)$ and a Groebner basis is $\{x - 1\}$.

Exercise 28.22: Euclidean algorithm giving the unit ideal

Write the two source generators as f, g. The successive remainder equations are

$$\begin{aligned} f &= (x^2+x)g + r_1, & r_1 &= x^2+3x-5,\\ g &= (x-4)r_1 + r_2, & r_2 &= 18x-21,\\ r_1 &= \left(\frac{x}{18}+\frac{25}{108}\right) r_2 - \frac{5}{36}. \end{aligned}$$

Each remainder lies in the ideal of the preceding pair, so the original ideal contains the nonzero constant $-5/36$, and therefore 1. The required Groebner basis is $\{1\}$.

Exercise 28.23: A two-point variety

Put $f = x^2y - x - 2$ and $g = xy + 2y - 9$. Then

$$f - (x-2)g = 4(2x + y - 5).$$

Thus $h = 2x+y-5$ lies in the ideal; furthermore $g-(y/2)h = -(y^2-9y+18)/2$. Set $k = y^2-9y+18$. These steps are reversible, so $(f, g) = (h, k)$. The leading monomials x and y^2 are relatively prime; the pair is a Groebner basis. Since $k = (y-3)(y-6)$, solve $y = 3$ or 6 and $x = (5-y)/2$. The real variety is $\{(1, 3), (-1/2, 6)\}$.

Exercise 28.24: A variety consisting of the origin

Let $f = x^2y + x$ and $g = xy^2 - y$. Their leading-term cancellation gives $yf - xg = 2xy$, so $h = xy$ lies in the ideal. Then $f - xh = x$ and $g - yh = -y$. Hence (f, g) contains both x, y. Conversely both f, g belong to (x, y), so the ideals are equal. Thus $\{x, y\}$ is a Groebner basis, and its common zero set is exactly $\{(0, 0)\}$.

Exercise 28.25: A cubic curve intersection reduced to one variable

For $f = x^2y + x + 1$ and $g = xy^2 + y - 1$, compute $yf - xg = x + y = h$. Dividing by h gives

$$g = y^2h - k, \qquad f = (xy - y^2 + 1)h + k, \qquad k = y^3 - y + 1.$$

These identities show $(f, g) = (h, k)$. The leading monomials x and y^3 are relatively prime, so $\{x + y, y^3 - y + 1\}$ is a Groebner basis. The real variety is $\{(-t, t) : t^3 - t + 1 = 0\}$. This cubic has exactly one real zero: its local maximum and minimum, at $t = \mp 1/\sqrt{3}$, are both positive, while its value tends to $-\infty$ as $t \to -\infty$. Thus the description gives a single real point without requiring a radical expression.

Exercise 28.26: A line and an isolated point

Let $f = x^2y + xy^2$ and $g = xy - x$. Set $h = f - (x + y + 1)g = x^2 + x$. Thus $(f, g) = (h, g)$. For lex order, their leading monomials are x^2 and xy. The only S-polynomial is

$$yh - xg = x^2 + xy = h + g,$$

which reduces to zero. Hence $\{x^2 + x, xy - x\}$ is a Groebner basis. The equations are $x(x + 1) = 0$ and $x(y - 1) = 0$. For $x = 0$, any y works; for $x = -1$, we must have $y = 1$. The variety is the entire y-axis together with the point $(-1, 1)$.

Exercise 28.27: True and false statements

(a) True. Here the book uses $F[\mathbf{x}] = F[x_1, \ldots, x_n]$. The Hilbert Basis Theorem gives a finite generating set for every ideal. This is not a claim that a multivariable ideal is principal.

(b) False. For example $\mathbb{R}^2 \setminus \{(0, 0)\}$ is not the zero set of polynomials: zero sets are closed, whereas this set is not.

(c) True. It is the zero set of the constant polynomial 1.

(d) True. A point (a, b) is the zero set of $(x - a)^2 + (y - b)^2$; a finite union of points is the zero set of the product of these polynomials.

(e) True. A line is the zero set of $ax + by + c$ with $(a, b) \neq (0, 0)$.

(f) True. A finite union of lines is the zero set of the product of their defining linear polynomials.

(g) True. The Euclidean algorithm preserves common divisors at each remainder step and terminates as degrees decrease.

(h) This depends on the reader's experience. If one has used the Euclidean algorithm in one variable or row reduction for linear polynomials, one has already computed special cases of Groebner bases; it is not a universal mathematical assertion about every reader.

(i) False. For example $\{x\}$ and $\{2x\}$ are distinct Groebner bases of (x). A *reduced monic* Groebner basis for a fixed monomial order is unique.

(j) False. $x \in (x, y)$ but $x \notin (x^2, y^2)$, since the latter ideal has no terms of degree one. Equal zero sets need not give equal ideals.

Exercise 28.28: Order is not divisibility

With $y < x$ in lex order, $y < x$ because the x exponent of y is zero and that of x is one. Yet y does not divide x: a product of y with a monomial must still have positive y exponent. This is the required counterexample.

Exercise 28.29: The ideal generated by a finite list

Source correction. The displayed coefficient condition $c_i \in I$ is circular; the intended condition is $c_i \in R$. With that correction define $I = \{\sum_{i=1}^{r} c_i f_i : c_i \in R\}$. Zero is obtained by all coefficients zero. A difference is obtained by subtracting the coefficients, so I is an additive subgroup. For $a \in R$,

$$a \sum_i c_i f_i = \sum_i (a c_i) f_i \in I.$$

Commutativity gives absorption on the right as well, proving the ideal claim. Unity ensures each f_i belongs to I, so this is exactly the smallest ideal containing the list.

Exercise 28.30: Common divisors survive a remainder step

If $d \mid f$ and $d \mid g$, then $d \mid (f - gq) = r$. Conversely if $d \mid g$ and $d \mid r$, then $d \mid (gq + r) = f$. Thus the common divisors of f, g are exactly those of g, r. This elementary equivalence is the reason the Euclidean algorithm computes a gcd.

Exercise 28.31: Checking a simple Groebner basis

For $f = xy$ and $g = y^2 - y$, the leading monomials under lex order $y < x$ are xy and y^2. Their least common multiple is xy^2, and

$$S(f, g) = yf - xg = xy = f.$$

It reduces to zero by subtracting f. Buchberger's criterion therefore proves that $\{xy, y^2 - y\}$ is a Groebner basis of the ideal it generates.

Exercise 28.32: The vanishing ideal of a set

The zero polynomial vanishes on S. If f, g vanish on S, then $(f - g)(s) = 0$ for every $s \in S$. If h is any polynomial, $(hf)(s) = h(s)f(s) = 0$ for every $s \in S$. These are precisely closure under differences and absorption, so $I(S)$ is an ideal in $F[x_1, \ldots, x_n]$.

Exercise 28.33: A set lies in its double construction

Take $s \in S$. By definition every $f \in I(S)$ vanishes at s. Hence s is a common zero of all polynomials in $I(S)$, which says $s \in V(I(S))$. Since s was arbitrary, $S \subset V(I(S))$.

Exercise 28.34: The double construction can enlarge a set

Take $S = \{(t, 0) : t \in \mathbb{R},\ t \neq 0\}$. If f vanishes on S, the one-variable polynomial $f(x, 0)$ has infinitely many roots and is therefore zero. Thus f is divisible by y, so $I(S) = (y)$. Consequently $V(I(S))$ is the full x-axis and contains $(0, 0)$, which is absent from S.

Exercise 28.35: An ideal lies in its double construction

If $f \in N$ and $s \in V(N)$, the definition of $V(N)$ gives $f(s) = 0$. Therefore f vanishes on every point of $V(N)$ and lies in $I(V(N))$. Thus $N \subset I(V(N))$. If $V(N)$ is empty, every polynomial vacuously vanishes on it and the inclusion still holds.

Exercise 28.36: The double construction can enlarge an ideal

Choose $N = (x^2, y^2) \subset \mathbb{R}[x, y]$. Its zero set is $\{(0, 0)\}$, so x vanishes on $V(N)$ and belongs to $I(V(N))$. But $x \notin N$: every term of every polynomial $ax^2 + by^2$ has total degree at least two. Hence $I(V(N)) \neq N$. In fact $I(V(N)) = (x, y)$.